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# A nonlinear dynamical approach to the path correction of multi-steering articulated vehicles

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## ABSTRACT

When designing articulated tractor-trailer systems, engineers should take care to contrast some undesired phenomena associated with the lateral dynamics: the path off-tracking, namely the trajectory deviation of the points of the two units compared to the midpoint of the tractor fore; the rearward amplification of the transverse acceleration. The off-tracking may lead the vehicle tail to invade the opposite direction lane or the emergency lane, whereas the rearward amplification may involve the danger of the trailer rollover. Nonetheless, the arrangement of multi-steering axle systems with the proper correlation among the steering angles may substantially cancel the off-tracking, both along a path with constant curvature and while entering or leaving the road bend. Moreover, the proper control of the increase and decrease of the steering angle during the transient phases can restrain the lateral acceleration of the towed unit. The present analysis considers several nonlinear effects of the vehicle lateral dynamics, such as large steering angles, nonlinear cornering characteristics and changes of the cornering forces because of the centrifugal effect. The goal is a practical formulation of the correlation among the steering angles to correct the vehicle run and ensure lateral stability.

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## 1. Introduction

A typical problem of the long vehicles is the off-tracking when undertaking the road bends or changing the lane. Actually, the different paths of their axle may involve the invasion of the opposite direction lane or the emergency lane, with the possible striking with fixed objects or walking people that may be present on the sides. The low-speed off-tracking appears as a sort of reluctance of the trailer tail to follow the tractor path, so that it remains inside the curve, whereas the high-speed off-tracking is due to the centrifugal effect, which tends to displace the vehicle tail outwards.

The transverse vehicle motions have been widely treated until now. They are associated with the lateral deformation of the tires and the consequent wheel sideslip. Rocard was maybe the first who stated a correlation between the wheel drift velocity and the transverse force at the ground print, defining the cornering stiffness of the tire, to be considered

an increasing function of the vertical load on the wheel [1]. More recently, Gillespie collected many experimental results about the influence of several geometrical and physical characteristics of the tires, such as inflation pressure, ply composition or aspect ratio [2]. A complete functional correlation between the lateral force and the sideslip angle was proposed by Pacejka with his 'magic formula', which permits adjusting several parameters to fit all types of tire characteristics [3].

As regards the lateral stability, innovative connection arrangements have been devised in the past for multi-trailer lorries, e. g. by double-bar linking of the tractor with an intermediate dolly, which moves the instant centre of relative rotation forward [4]. Moreover, special control systems may restore the correct run: differential braking of the left and right wheels to induce corrective torques; steer-by-wire devices, which correct the steering angle input; active differential devices, which control the distribution of the drive torque among the wheels [5–10]. Also, oscillation dampers might be applied, such as dynamic absorbers or gyrostabilisers, whose action is similar to the anti-roll devices in use on some transatlantic liners of the past (e. g. see [11] for the naval application).

The main object of the present study is the path optimisation of an articulated vehicle when undertaking generic direction changes, for example when entering or leaving a road bend or during the lane changes. For this purpose, a five-axle vehicle train composed of a tractor and a semitrailer will be here considered, where all axles will be assumed to steer, and the best correlation among the steering angles will be sought to cancel undesired path deviations at low speeds and limit the lateral acceleration peaks at high speed.

The multiple-axle steering technique has recently taken on in the high-class automotive area and the field of articulated vehicles and movable farm machines, aiming at better manoeuvring in restricted spaces and at limiting the tire skidding and wearing. To achieve this, complex electro-hydraulic control systems have been devised to steer the wheels so that the geometrical axes converge to the vehicle rotation centre. We here cite only some papers on this topic. Bushnell et al. consider three-input non-holonomic systems and give simulation results [12]. Wu and Hai address the yaw and roll motion of a multi-axle articulated vehicle [13,14]. Sathyabalam et al. propose multiple mode steering systems with mechanical commutation [15]. Aoki et al. analyse the lateral dynamics of multi-articulated vehicles with an optional number of trailers [16]. Wagner et al. study articulated three-unit buses with kinematic steering control [17,18]. Du et al. study electro-hydraulic servo-systems for the kinematical steering of multi-axle vehicles [19]. Cheng et al. consider the active steering of the trailers to cancel off-tracking [20,21]. Wang et al. apply the multi-axle active steering to control the lateral acceleration of the rearward trailer [22]. Wang and He focus on the rearward amplification of multi-trailer vehicles with active steering systems [23]. Kharrazi et al. study active steering systems for reducing the yaw rearward amplification and the off-tracking [24]. Islam et al. consider an active steering system to reduce high speed lateral stability and low speed off-tracking [25]. Jujnovich et al. consider control systems for multi-steering articulated vehicles and present the results of numerical simulations and experimental tests on the road [26,27]. Kural et al. formulate a path following approach and focus on the low-speed manoeuvrability and high-speed stability [28].

All these approaches offer valuable solutions to the problem of the path optimisation of a long vehicle. Nevertheless, it may be opportune and cheaper to reach an acceptable path correction by a simple and reliable steering strategy, based on precise correlation laws among the steering angles applicable through an open-loop conception, not subject to

possible dangerous failures of feedback lines or sensors. This benefit should result from the feedforward structure without sophisticated components of the steering control and, applying proper values of the parameters, acceptable accuracy should be expected by the road tests comparing with more elaborate systems. The fundamental requirement for the articulated trucks will be here the off-track suppression by a simple steering system with a long operative life. In parallel, particular care will be taken of the transient phases to limit possible acceleration peaks and ensure rollover stability.

The motion equations will be firstly solved by numerical procedures, changing the steering angles iteratively with the aim at their best correlation with the driver steering angle and the vehicle speed. Then, suitable formulae matching the numerical results will be developed, aiming at the real steering control en route.

Due to the largeness of the steering angles and the sideslip angles in some conditions under examination (slow speed manoeuvring in restricted spaces) and to the remarkable changes of the lateral cornering forces due to the centrifugal effect at high speed, the formulation of the motion equations will be here nonlinear.

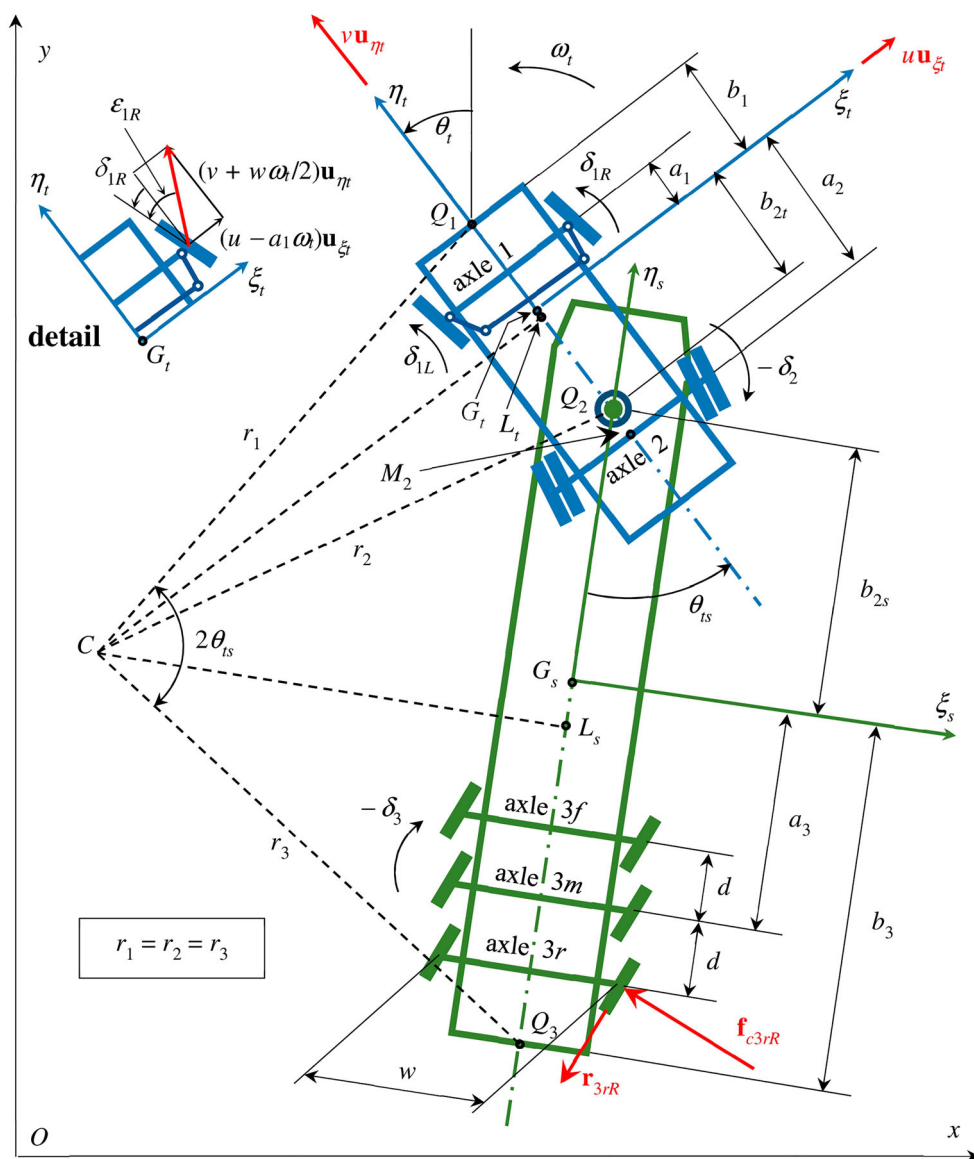
## 2. Mathematical model

### 2.1. Scheme of the articulated vehicle

The tractor-semitrailer configuration under examination is very common in the European roads and is qualitatively shown in Figure 1, whereas the real data are those of Table 1. The frames  $Oxyz$ ,  $G_t\xi_t\eta_t\zeta_t$  and  $G_s\xi_s\eta_s\zeta_s$  are fixed to the ground, the tractor and the semitrailer, respectively;  $\theta_t$  and  $\theta_{ts}$  are the angles that the  $\eta_t$  axis of the tractor forms with the  $y$ -axis and with the  $\eta_s$  axis, respectively;  $G_t$ ,  $m_t$ ,  $\rho_t$ , and  $G_s$ ,  $m_s$ ,  $\rho_s$ , are the barycentres, the masses and the barycentric gyration radii of the tractor and the semi-trailer;  $Q_2$  is the centre of the fifth wheel coupling;  $\delta_{1L}$  and  $\delta_{1R}$  are the steering angles of the left and right wheels of the first axle;  $\delta_2$  and  $\delta_3$  are the steering angles of the rear axle of the tractor and of the tridem axles of the semitrailer, which are both supposed equal on the left and right sides, while  $\delta_3$  is also supposed equal in the fore (3f), mid (3m) and rear (3r) axles;  $v$  is the vehicle speed. All angles and angular speeds are positive when counter-clockwise directed.

It is supposed that:

- a conventional four-bar device interfaces the driver with the fore steering wheels, whose rotation angles,  $\delta_{1L}$  and  $\delta_{1R}$ , are readily calculable as functions of their average value,  $\delta_1 = (\delta_{1L} + \delta_{1R})/2$  by the well-known closure formulae of the four-bar device, whence  $\delta_1$  may be chosen as an input variable;
- the vehicle train runs on a level road, where the rolling ground resistances,  $\mathbf{r}_{ij}$ , are parallel to the wheel planes and have magnitude  $r_{ij}$  on the single wheel of the axle  $i$  ( $= 1, 2, 3f, 3m, 3r$ ) on the side  $J$  ( $= L$ , left, or  $R$ , right); here,  $r_{ij} = f_r l_{ij}$ , where  $f_r$  is a constant coefficient of rolling friction and  $l_{ij}$  is the vertical load on the wheel;
- the suspension stiffness is sufficiently large and the possible vehicle motions on the vertical planes do not interfere in the lateral motion;
- the air resistances have magnitude  $a_t$  and  $a_s$  on the tractor and the semitrailer and are significant only for appreciable vehicle speeds and then for rather large path radii; hence, their directions are considered approximately parallel to the longitudinal axes of the



**Figure 1.** Scheme of the tractor-semitrailer system. The ground forces are shown for the last wheel on the right only ( $\mathbf{f}_{c3rR}$  and  $\mathbf{r}_{3rR}$ ). The wheel track  $w$  is the same for all axles. The angular quantities are positive when counterclockwise directed, whence  $\delta_2 < 0$ ,  $\delta_3 < 0$  in the particular configuration of this figure. The detail justifies the expression (10) of  $\varepsilon_{1R}$ . The other sideslip angles are calculable likewise.

two units and  $a_i$  and  $a_s$  are expressed in the form  $c_x \rho S v^2 / 2$ , where  $c_x$  is the aerodynamic coefficient,  $\rho$  is the air mass density and  $S$  is the frontal section;

- the driving wheels are in dual combination on the rear axle 2 of the tractor; the total tractive ground force is parallel to the wheel planes, includes the rolling resistances as subtractive terms and has magnitude  $t_2$ ; the shares of  $t_2$  among the four wheel are equal, on the left and right sides because of the differential gear;

**Table 1.** Data of example case (taken from real vehicles and typical operative conditions).

|                   |        |            |                                                  |
|-------------------|--------|------------|--------------------------------------------------|
| $a_1$             | 1 m    | $a_s/v^2$  | $0.0033 \text{ kN} \times \text{s}^2/\text{m}^2$ |
| $a_2$             | 2.6 m  | $a_t/v^2$  | $0.0045 \text{ kN} \times \text{s}^2/\text{m}^2$ |
| $a_3$             | 3.6 m  | $c'$       | $253.675 \text{ kN/rad}$                         |
| $b_1$             | 2 m    | $c''$      | $92.810 \text{ kN/rad}$                          |
| $b_{2t}$          | 2.1 m  | $f_r$      | 0.012                                            |
| $b_{2s}$          | 5.4 m  | $m_t g$    | 80 kN                                            |
| $b_3$             | 6.25 m | $m_s g$    | 400 kN                                           |
| $d$               | 1.31 m | $\rho_t^2$ | $3.5 \text{ m}^2$                                |
| $w$               | 2 m    | $\rho_s^2$ | $13.5 \text{ m}^2$                               |
| $h_{Gt} = h_{Gs}$ | 2 m    |            |                                                  |

- the preliminary connecting operation between the tractor and the semitrailer is carried out by lifting the fifth wheel plane until achieving the horizontality of the semitrailer loading bed, which implies the same compression of all suspension springs of the tri-dem set; therefore, all vertical loads on the semitrailer wheels are initially equal but may change with the load and the centrifugal force.

## 2.2. Vehicle kinematics

Indicate with  $u$  and  $v$  the velocity components of  $G_t$  along the transverse and longitudinal axes  $\xi_t$  and  $\eta_t$ , where  $v$  is supposed constant and much larger than  $u$ . Moreover, define the angular velocities,  $\omega_t = d\theta_t/dt$  of the tractor and  $\omega_{ts} = d\theta_{ts}/dt$  of the tractor relative to the trailer, where  $t$  is the time.

Using bold types for vectors and introducing the unit vectors,  $\mathbf{u}_{\xi_t}$ ,  $\mathbf{u}_{\eta_t}$ ,  $\mathbf{u}_{\zeta_t}$ ,  $\mathbf{u}_{\xi_s}$ ,  $\mathbf{u}_{\eta_s}$ ,  $\mathbf{u}_{\zeta_s}$ , parallel to the axes of the two moving frames  $G_t\xi_t\eta_t\zeta_t$  and  $G_s\xi_s\eta_s\zeta_s$ , the velocity vectors of the barycentres  $G_t$  and  $G_s$  are

$$\mathbf{v}_{Gt} = u\mathbf{u}_{\xi_t} + v\mathbf{u}_{\eta_t} \quad (1)$$

$$\begin{aligned} \mathbf{v}_{Gs} = & [(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + b_{2s}(\omega_t - \omega_{ts})] \mathbf{u}_{\xi_s} \\ & + [(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts}] \mathbf{u}_{\eta_s} \end{aligned} \quad (2)$$

and, as  $d\mathbf{u}_{\xi_t}/dt = \omega_t \mathbf{u}_{\eta_t}$ ,  $d\mathbf{u}_{\eta_t}/dt = -\omega_t \mathbf{u}_{\xi_t}$ ,  $d\mathbf{u}_{\xi_s}/dt = (\omega_t - \omega_{ts}) \mathbf{u}_{\eta_s}$ ,  $d\mathbf{u}_{\eta_s}/dt = -(\omega_t - \omega_{ts}) \mathbf{u}_{\xi_s}$  by Poisson's formulas, the acceleration vectors are

$$\mathbf{a}_{Gt} = \left( \frac{du}{dt} - \omega_t v \right) \mathbf{u}_{\xi_t} + u\omega_t \mathbf{u}_{\eta_t} \quad (3)$$

$$\begin{aligned} \mathbf{a}_{Gs} = & \left[ \left( \frac{du}{dt} + b_{2t} \frac{d\omega_t}{dt} \right) \cos \theta_{ts} - (u + b_{2t}\omega_t) \omega_t \sin \theta_{ts} \right. \\ & \left. - v\omega_t \cos \theta_{ts} + b_{2s} \left( \frac{d\omega_t}{dt} - \frac{d\omega_{ts}}{dt} \right) \right] \mathbf{u}_{\xi_s} \\ & + \left[ \left( \frac{du}{dt} + b_{2t} \frac{d\omega_t}{dt} \right) \sin \theta_{ts} + (u + b_{2t}\omega_t) \omega_t \cos \theta_{ts} \right. \\ & \left. - v\omega_t \sin \theta_{ts} + b_{2s}(\omega_t - \omega_{ts})^2 \right] \mathbf{u}_{\eta_s} \end{aligned} \quad (4)$$

whence the inertia forces can be readily calculated.

### 2.3. Tire cornering

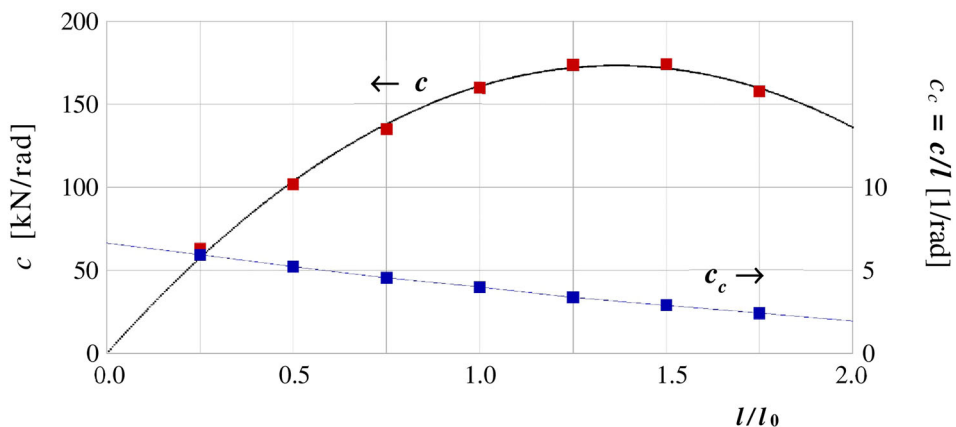
The cornering ground force acting orthogonally to the wheel plane is correlated with the sideslip angle but depends also on other working conditions, such as the vertical load  $l$ , the inflation pressure, the ply composition and the tire aspect ratio. Several formulae have been proposed in the literature to consider the many aspects of the cornering phenomenon. Here, only the variability with the sideslip angle and with the vertical load will be considered.

Simplifying Pacejka's formula [3] where the magnitude  $f_c$  of the cornering force  $\mathbf{f}_c$  is defined by an odd function of the sideslip angle  $\varepsilon$ , an approximate formulation will be introduced assuming the odd cubic law

$$f_c \cong -c(\varepsilon - p_3 \varepsilon^3) \quad (5)$$

where  $c$  is the cornering stiffness for  $\varepsilon \rightarrow 0$  and the number  $p_3$  is such that  $df_c/d\varepsilon$  vanishes for a certain value of  $\varepsilon$ , say  $\varepsilon_{\max.}$ , where  $|f_c|$  reaches its maximum value  $f_{c\max.}$  of incipient gross slip. Thus,  $p_3 \varepsilon_{\max.}^2 = 1/3$  and  $\varepsilon_{\max.} = 1.5 f_{c\max.}/c$ .

Reference [2] also defines the cornering coefficient  $c_c = c/l$  (cornering stiffness  $c$ , normalised by vertical load  $l$ ) and shows in Figure 10.15 that  $c_c$  is a nearly linear decreasing function of the load fraction  $l/l_0$ , where  $l_0$  is the nominal vertical load on the wheel, under the design conditions of Table 1 in the present case. These conditions were so fixed to give equal sharing of the total load among the wheels, whence  $l_0 = m_{\text{tot.}}g/n_{\text{tot.}}$ , where  $g$  is the gravity acceleration,  $m_{\text{tot.}}$  and  $n_{\text{tot.}}$  are the total mass and the total wheel number of the vehicle train. Figure 2 shows the plots of  $c$  and  $c_c$  and is similar to that reported by [2] for a passenger car (Figure 10.15), save proper scale factors for the ordinates, which consider the different weight and cornering stiffness of the present articulated vehicle. These factors were chosen to get  $c = 160$  kN and  $c_c = 4$  for all wheels at rated load ( $l/l_0 = 1$ ), which values are quite realistic for a typical truck system.



**Figure 2.** Cornering stiffness  $c$  and cornering coefficient  $c_c$  versus vertical load. Dots refer to Figure 10.15 of reference [2] save different scale factors for the ordinates, which apply to lorries instead of passenger cars.

Then, since  $d^2(c/l)/dl^2 \approx 0$ , the following approximate function  $c(l/l_0)$  was assumed

$$c = c' \left( \frac{l}{l_0} \right) - c'' \left( \frac{l}{l_0} \right)^2 \quad (6)$$

and the coefficients  $c'$  and  $c''$  were calculated through a process of mean square error minimisation, obtaining  $c' = 253.675 \text{ kN}$  and  $c'' = 92.810 \text{ kN}$ .

On changing the freight on the semitrailer in comparison with the design conditions or the freight distribution on the loading bed, all the ratios  $l/l_0$  change (static changes:  $l_{\text{stat.}}/l_0 \neq 1$ ). Furthermore, the centrifugal effect along the road bends increases the vertical load on the external wheels and decreases it on the internal ones (dynamic changes:  $l_{\text{external}} > l_{\text{internal}}$ ), whence the cornering forces depend not only on the freight but also on the running condition. The moment of the centrifugal force tending to overturn each vehicle unit is  $ma_G h_G$ , where  $a_G$  and  $h_G$  are the transverse acceleration and the height of the barycentre  $G$ . Considering the total changes of the vertical loads on the wheels, one has

$$\frac{l}{l_0} = \frac{l_{\text{stat.}}}{l_0} \left( 1 \pm \frac{2a_G h_G}{gw} \right) \quad (7)$$

where  $w$  is the wheel track and the plus and minus signs refer to the external and internal wheels.

Therefore, using Eqs. (6–7), the cornering stiffness of the wheels on each side of an axle is

$$c_e = n_w \left[ c' \left( \frac{l_{\text{stat.}}}{l_0} \right) \left( 1 + \frac{2a_G h_G}{gw} \right) - c'' \left( \frac{l_{\text{stat.}}}{l_0} \right)^2 \left( 1 + \frac{2a_G h_G}{gw} \right)^2 \right] \quad (8)$$

$$c_i = n_w \left[ c' \left( \frac{l_{\text{stat.}}}{l_0} \right) \left( 1 - \frac{2a_G h_G}{gw} \right) - c'' \left( \frac{l_{\text{stat.}}}{l_0} \right)^2 \left( 1 - \frac{2a_G h_G}{gw} \right)^2 \right] \quad (9)$$

where the subscripts  $e$  and  $i$  refer to the  $c$  (or sums of  $c$ ) of the external and internal wheels of the axle,  $n_w$  is the wheel number on each side (1, or else 2 for dual wheels) and one must check that the ratio  $2a_G h_G/(gw)$  is sufficiently lower than 1 to prevent the vehicle roll-over. In practice, the centrifugal effect decreases the total cornering force of each axle somehow.

The sideslip angle  $\varepsilon$  of each wheel is the angle formed by the velocity vector of the wheel centre with the rim plane. The velocity components of the centres of the right and left wheels of each axle along the axle direction are equal to the analogous velocity component of the axle midpoint due to the axle rigidity, whereas the components orthogonal to the axle of the vehicle differ due to the vehicle rotation.

The sideslip angles of the wheels of axles 1 and 2 are, by Figure 1,

$$\varepsilon_{1R} = \arctan \left( \frac{u - a_1 \omega_t}{v + w \omega_t / 2} \right) + \delta_{1R} \quad (10)$$

$$\varepsilon_{1L} = \arctan \left( \frac{u - a_1 \omega_t}{v - w \omega_t / 2} \right) + \delta_{1L} \quad (11)$$

$$\varepsilon_{2R} = \arctan \left( \frac{u + a_2 \omega_t}{v + w \omega_t / 2} \right) + \delta_2 \quad (12)$$



$$\varepsilon_{2L} = \arctan\left(\frac{u + a_2\omega_t}{v - w\omega_t/2}\right) + \delta_2 \quad (13)$$

Indicating the axle inter-distance of the tridem set with  $d$  and observing Figure 1, the sideslip angles of the wheels of the axles  $3f$ ,  $3m$ ,  $3r$  may be written in the form

$$\varepsilon_{3fR} = \arctan\left[\frac{(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + (\omega_t - \omega_{ts})(b_{2s} + a_3 - d)}{(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts} + (\omega_t - \omega_{ts})w/2}\right] + \delta_3 \quad (14)$$

$$\varepsilon_{3fL} = \arctan\left[\frac{(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + (\omega_t - \omega_{ts})(b_{2s} + a_3 - d)}{(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts} - (\omega_t - \omega_{ts})w/2}\right] + \delta_3 \quad (15)$$

$$\varepsilon_{3mR} = \arctan\left[\frac{(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + (\omega_t - \omega_{ts})(b_{2s} + a_3)}{(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts} + (\omega_t - \omega_{ts})w/2}\right] + \delta_3 \quad (16)$$

$$\varepsilon_{3mL} = \arctan\left[\frac{(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + (\omega_t - \omega_{ts})(b_{2s} + a_3)}{(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts} - (\omega_t - \omega_{ts})w/2}\right] + \delta_3 \quad (17)$$

$$\varepsilon_{3rR} = \arctan\left[\frac{(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + (\omega_t - \omega_{ts})(b_{2s} + a_3 + d)}{(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts} + (\omega_t - \omega_{ts})w/2}\right] + \delta_3 \quad (18)$$

$$\varepsilon_{3rL} = \arctan\left[\frac{(u + b_{2t}\omega_t) \cos \theta_{ts} - v \sin \theta_{ts} + (\omega_t - \omega_{ts})(b_{2s} + a_3 + d)}{(u + b_{2t}\omega_t) \sin \theta_{ts} + v \cos \theta_{ts} - (\omega_t - \omega_{ts})w/2}\right] + \delta_3 \quad (19)$$

Then, the magnitude of the cornering forces may be calculated using Eqs. (5) to (19) and may be briefly indicated with the symbols  $f_{cij}$ , where  $i = 1, 2, 3f, 3m, 3r$ , and  $j = R, L$ .

## 2.4. Equations of motion

A distinction must be made first between the ‘intrinsic’ vehicle dynamics, which considers the vehicle equilibrium independently of the fixed reference frame  $Oxyz$  and applies to the steady equilibrium and to the motion stability, and the vehicle dynamics in the external environment, which applies to the comparison of the vehicle point trajectories in transient conditions and adds proper kinematical equations to the intrinsic equilibrium equations.

Indicating the total magnitude of the cornering forces and the rolling resistances on the right and left sides of the tridem set with  $f_{c3R}$ ,  $f_{c3L}$  and  $r_{3R}$ ,  $r_{3L}$  for brevity, the intrinsic equations of motion may be obtained by the classical Newton approach and may be written in the convenient form

$$\begin{aligned} & [m_s(b_{2t} - a_2) - m_t a_2] \frac{du}{dt} + [m_s(b_{2t} - a_2)(b_{2t} + b_{2s} \cos \theta_{ts}) + m_t \rho_t^2] \frac{d\omega_t}{dt} \\ & - m_s(b_{2t} - a_2) b_{2s} \cos \theta_{ts} \frac{d\omega_{ts}}{dt} \\ & = [m_s(b_{2t} - a_2) - m_t a_2] v \omega_t - m_s(b_{2t} - a_2) b_{2s} \sin \theta_{ts} (\omega_t - \omega_{ts})^2 \\ & + (f_{c1R} \cos \delta_{1R} + f_{c1L} \cos \delta_{1L} - r_{1R} \sin \delta_{1R} - r_{1L} \sin \delta_{1L})(a_1 + a_2) \\ & - \frac{w}{2} [f_{c1R} \sin \delta_{1R} - f_{c1L} \sin \delta_{1L} + r_{1R} \cos \delta_{1R} - r_{1L} \cos \delta_{1L} + (f_{c2R} - f_{c2L}) \sin \delta_2] \\ & - (b_{2t} - a_2) [(f_{c3R} + f_{c3L}) \cos(\theta_{ts} - \delta_3) + (r_{3R} + r_{3L}) \sin(\theta_{ts} - \delta_3) + a_s \sin \theta_{ts}] \end{aligned} \quad (20)$$

$$\begin{aligned}
& m_s b_{2s} \cos \theta_{ts} \frac{du}{dt} + [m_s \rho_s^2 + m_s b_{2s} (b_{2s} + b_{2t} \cos \theta_{ts})] \frac{d\omega_t}{dt} - m_s (\rho_s^2 + b_{2s}^2) \frac{d\omega_{ts}}{dt} \\
& = m_s b_{2s} [v\omega_t \cos \theta_{ts} + \omega_t (u + b_{2t} \omega_t) \sin \theta_{ts}] \\
& \quad - [(f_{c3R} + f_{c3L}) \cos \delta_3 - (r_{3R} + r_{3L}) \sin \delta_3] (b_{2s} + a_3) \\
& \quad - \frac{w}{2} [(f_{c3R} - f_{c3L}) \sin \delta_3 + (r_{3R} - r_{3L}) \cos \delta_3] \\
& \quad - (f_{c3rR} + f_{c3rL} - f_{c3fR} - f_{c3fL}) d \cos \delta_3
\end{aligned} \tag{21}$$

$$\begin{aligned}
& (m_t + m_s) \cos \delta_2 \frac{du}{dt} + [m_s b_{2t} \cos \delta_2 + m_s b_{2s} \cos(\theta_{ts} + \delta_2)] \frac{d\omega_t}{dt} \\
& \quad - m_s b_{2s} \cos(\theta_{ts} + \delta_2) \frac{d\omega_{ts}}{dt} \\
& = (m_t + m_s) v\omega_t \cos \delta_2 - [m_t u \omega_t + m_s (u + b_{2t} \omega_t) \omega_t] \sin \delta_2 \\
& \quad - m_s b_{2s} (\omega_t - \omega_{ts})^2 \sin(\theta_{ts} + \delta_2) - a_s \sin(\theta_{ts} + \delta_2) - a_t \sin \delta_2 \\
& \quad - f_{c1R} \cos(\delta_{1R} - \delta_2) - f_{c1L} \cos(\delta_{1L} - \delta_2) \\
& \quad + r_{1R} \sin(\delta_{1R} - \delta_2) + r_{1L} \sin(\delta_{1L} - \delta_2) - f_{c2R} - f_{c2L} \\
& \quad - (f_{c3R} + f_{c3L}) \cos(\theta_{ts} - \delta_3 + \delta_2) - (r_{3R} + r_{3L}) \sin(\theta_{ts} - \delta_3 + \delta_2)
\end{aligned} \tag{22}$$

$$\frac{d\theta_{ts}}{dt} = \omega_{ts} \tag{23}$$

where the time derivatives are separated on the left sides, which are linear in these derivatives. We have, in matrix form,

$$\mathbf{M} \frac{d\mathbf{q}}{dt} = \mathbf{n}(\mathbf{q}) \tag{24}$$

where  $\mathbf{q}$  is the state vector  $\{u, \omega_t, \omega_{ts}, \theta_{ts}\}^T$ , the matrix  $\mathbf{M}$  is formed by the factors of the components of  $d\mathbf{q}/dt$  at the left side of Eqs. (20–23) and the vector  $\mathbf{n}$  is formed with the right sides. Pre-multiplying both sides of Eq. (24) by  $\mathbf{M}^{-1}$ , the differential system gets ready for the numerical integration that will be carried out for the transient conditions in Section 4.

Equation (20) gives the rotational equilibrium of the tractor around the midpoint  $M_2$  of axle 2, Equation (21) gives the rotational equilibrium of the semitrailer around the centre  $Q_2$  of the fifth wheel coupling and Equation (22) gives the translational equilibrium of the whole vehicle along the direction orthogonal to the wheels of axle 2 (see Figure 1). This formulation eliminates the unknown tractive force  $t_2$ , which might be calculated by the longitudinal equilibrium condition if desired.

The equation system (20–23 or 24) do not contain the angle  $\theta_t$ , nor the coordinates  $x$  and  $y$  of any point of the vehicle, because the intrinsic dynamic is independent of the choice of the fixed frame.

As regards the lateral stability, it is here worth noting that the configuration of Figure 1 does not reveal problems in general, excepting possible oscillating instability conditions for abnormal load distribution, e. g. large loads on the back of the loading bed of the semi-trailer. This can be ascertained by numerical integration of the above nonlinear equations after imposing casual initial disturbances or by the solution of the eigenvalue problem of the linearised equations in the hypothesis of small perturbations.

### 3. Steady off-tracking elimination

Assume that, undertaking a road bend with constant curvature, the driver imposes an initial steering command by an input  $\delta_1$  and a final one by another input  $-\delta_1$  to return to the straight path, thus inducing two transient periods, of entry and exit, and an in-between steady state.

We will only consider the steady run in this section, where all the vehicle points run along concentric circular paths, and will simply call  $\omega$  the common angular speed of the tractor and the semitrailer around the centre. We will also look for proper functions  $\delta_2(\delta_1, \nu)$  and  $\delta_3(\delta_1, \nu)$  to cancel the vehicle off-tracking for various operative conditions.

It is possible to assume the differences among the path radii of three convenient control points on the longitudinal axes of the vehicle as off-tracking indicators, for example, the midpoint  $Q_1$  of the tractor front end, the centre  $Q_2$  of the fifth wheel coupling and the midpoint  $Q_3$  of the semitrailer rear end.

The transverse velocity is equal to  $u - \eta_t \omega$  along the longitudinal axis of the tractor and vanishes at a certain point  $L_t$  where  $\eta_t(L_t) = u/\omega$ , whence the centre  $C_t$  of the circular paths of  $L_t$  and all the other tractor points lies on the normal to the  $\eta_t$  axis drawn from  $L_t$  and has transverse coordinate  $\xi_t(C_t) = -\nu/\omega$ . Therefore, the trajectories of  $Q_1$  and  $Q_2$  have radii  $r_1 = \sqrt{\xi_t^2(C_t) + [b_1 - \eta_t(L_t)]^2}$  and  $r_{2t} = \sqrt{\xi_t^2(C_t) + [b_{2t} + \eta_t(L_t)]^2}$ , where  $b_1$  and  $b_{2t}$  are the distances of  $Q_1$  and  $Q_2$  from  $G_t$  (Figure 1). In order that the control points  $Q_1$  and  $Q_2$  have the same path radii ( $r_1 = r_{2t}$ ), it must be  $2\eta_t(L_t) = b_1 - b_{2t}$  that is

$$\frac{u}{\omega} = \frac{b_1 - b_{2t}}{2} \quad (25)$$

Likewise, the transverse velocity vanishes at a certain point  $L_s$  of the longitudinal axis of the semitrailer where  $\eta_s(L_s) = b_{2s} + [(u + \omega b_{2t})\cos\theta_{ts} - \nu\sin\theta_{ts}]/\omega = 0$ , where the term inside square brackets is the component along the  $\xi_s$  axis of the velocity of  $Q_2$ . Hence the centre  $C_s$  of the circular paths of  $L_s$  and all the other points of the semitrailer lies on the normal to the  $\eta_s$  axis drawn from  $L_s$  and has transverse coordinate  $\xi_s(C_s) = -[(u + \omega b_{2t})\sin\theta_{ts} + \nu\cos\theta_{ts}]/\omega$ , where the term inside square brackets is the common velocity component along the  $\eta_s$  axis of all the semitrailer points lying on that axis. Then, the trajectories of  $Q_2$  and  $Q_3$  have radii  $r_{2s} = \sqrt{\xi_s^2(C_s) + [b_{2s} - \eta_s(L_s)]^2}$  and  $r_3 = \sqrt{\xi_s^2(C_s) + [b_3 + \eta_s(L_s)]^2}$ , where  $b_{2s}$  and  $b_3$  are the distances of  $Q_2$  and  $Q_3$  from  $G_s$  (Figure 1). Using the above expressions of  $\xi_t(C_t)$ ,  $\eta_t(L_t)$ ,  $\xi_s(C_s)$ ,  $\eta_s(L_s)$ , it is easy to realise that  $r_{2t} = r_{2s}$  and  $C_t \equiv C_s$  for the tractor and semitrailer, as is indeed to be expected, whence and it is possible to drop the subscripts  $t$  and  $s$  ( $r_2 = r_{2t} = r_{2s}$ ,  $C \equiv C_t \equiv C_s$ ). In order that  $Q_2$  and  $Q_3$  have the same path radii ( $r_2 = r_3$ ), it must be  $2\eta_s(L_s) = b_{2s} - b_3$  that is

$$\frac{\nu}{\omega} \sin\theta_{ts} - \frac{u}{\omega} \cos\theta_{ts} = \frac{b_{2s} + b_3}{2} + b_{2t} \cos\theta_{ts} \quad (26)$$

and replacing  $u/\omega$  from Eq. (25) into Eq. (26), one obtains

$$\omega = \frac{2\nu \sin\theta_{ts}}{b_{2s} + b_3 + (b_1 + b_{2t}) \cos\theta_{ts}} \quad (27)$$

Thus, imposing the path coincidence of the three control points  $Q_1$ ,  $Q_2$  and  $Q_3$ , Equation (27) gives the functional dependence  $\omega(\theta_{ts}, \nu)$ , whence Equation (25) permits obtaining  $u(\theta_{ts}, \nu)$ .

Fixing two values for  $\delta_1$  and  $\nu$ , cancelling all time derivatives in Eqs. (20–23) as we are analysing the steady run along a road curve, imposing  $\omega_{ts} = 0$  by Eq. (23) and replacing  $u$  and  $\omega_t (= \omega)$  by Eqs. (25) and (27), we get three nonlinear equations in the unknowns  $\delta_2$ ,  $\delta_3$  and  $\theta_{ts}$ , which can be solved by numerical methods to obtain the desired laws,  $\delta_2(\delta_1, \nu)$ ,  $\delta_3(\delta_1, \nu)$ ,  $\theta_{ts}(\delta_1, \nu)$ , that give the same paths of the three control points. In practice, we nest two iteration loops for the two steering angles,  $\delta_2$  and  $\delta_3$ , into a third one for  $\theta_{ts}$ , introducing suitable tactics to increase the precision when approaching the convergence. The first internal loop iterates  $\delta_3$  in the rotational equation of the semitrailer, Eq. (21), which does not contain  $\delta_2$ , then the second internal loop iterates  $\delta_2$  in the translational equation of the whole vehicle, Eq. (22), and finally, the remaining equation, Eq. (20), is used in the external loop for  $\theta_{ts}$ .

The ranges of the admissible values of the angles  $\delta_i$  are bounded by the limits defined in Section 2.3 for the corresponding  $\varepsilon_i$ :  $|\varepsilon_i| < \varepsilon_{\max}$ . The angle  $\theta_{ts}$ , which has the same sign of  $\delta_1$ , is limited by the maximum value  $\theta_{ts\max}$ , corresponding to the maximum path curvature of the control points  $Q_1$ ,  $Q_2$  and  $Q_3$ . Since  $Q_1$ ,  $Q_2$  and  $Q_3$  belong to the same circle of radius  $r$  and centre  $C$  in the hypothesis of correct steady paths and the angle  $\widehat{Q_1 Q_2 Q_3}$  is equal to  $\pi - \theta_{ts}$ , then the angle  $\widehat{Q_1 C Q_3}$  is equal to  $2\theta_{ts}$  and one gets by Carnot's theorem

$$2r \sin \theta_{ts} = \sqrt{(b_1 + b_{2t})^2 + (b_{2s} + b_3)^2 + 2(b_1 + b_{2t})(b_{2s} + b_3) \cos \theta_{ts}} \quad (28)$$

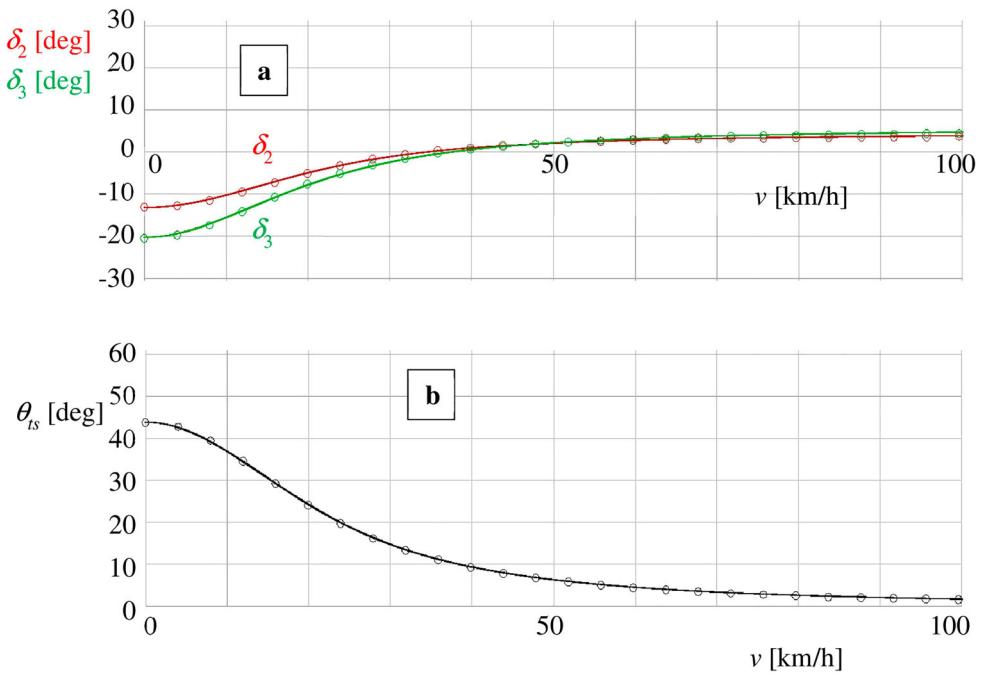
and then

$$\cos^2 \theta_{ts} + \frac{(b_1 + b_{2t})(b_{2s} + b_3)}{2r^2} \cos \theta_{ts} + \frac{(b_1 + b_{2t})^2 + (b_{2s} + b_3)^2}{4r^2} - 1 = 0 \quad (29)$$

Increasing the angle  $\theta_{ts}$ , the tractor axis rotates with respect to the semitrailer around  $Q_2$  and the centre  $C$  of the circumference through  $Q_1, Q_2, Q_3$  is always identified by the intersection of the axes of the two straight segments  $\overline{Q_1 Q_2}$  and  $\overline{Q_2 Q_3}$ , the longest of which is  $\overline{Q_2 Q_3}$ . Of course, the circle radius becomes minimum when  $C$  coincides with the midpoint of  $\overline{Q_2 Q_3}$  and  $2r = b_{2s} + b_3$  whence, replacing  $2r$  with  $b_{2s} + b_3$  in Eq. (29), one finds that the discriminant of this quadratic equation in  $\cos \theta_{ts}$  vanishes and  $\cos \theta_{ts\max} = -(b_1 + b_{2t})/(b_{2s} + b_3)$ , so that  $\theta_{ts\max} > 90^\circ$ .

The single iterative searches for  $\delta_2$ ,  $\delta_3$  and  $\theta_{ts}$  may be carried out starting from one end of the respective admissibility ranges and proceeding towards the other end until the corresponding equation is satisfied. At last, another limit to be verified *ex-post* implies checking that the ratios  $2a_{Gt}h_G/(gw)$  and  $2a_{Gs}h_G/(gw)$  are lower than one to prevent rollover (or else  $< 0.7$  assuming a safety factor equal to 0.7).

Figure 3(a,b) show the diagrams of the three angles  $\delta_2$ ,  $\delta_3$  and  $\theta_{ts}$  on varying the vehicle speed for a prefixed value of  $\delta_1$ . One observes that  $\delta_2$  and  $\delta_3$  are negative in the low speed range, where they must correct the reluctance of the articulated vehicle tail to follow the path of the tractor, and are positive in the high-speed range, where they must compensate the centrifugal effect. The circles on the plots of  $\delta_2$  and  $\delta_3$  refer to the matching analytical functions that will be reported later on in Section 5, whereas the circles on the plot of  $\theta_{ts}$



**Figure 3.** Optimised steady run with zero off-tracking along a constant curvature path ( $\delta_1 = 6$  deg). (a) steering angles  $\delta_2$  and  $\delta_3$  versus vehicle speed; circles refer to analytical approximation by Eqs. (45–51); (b) angle  $\theta_{ts}$  between the two units versus vehicle speed; circles refer to analytical approximation by Eqs. (30–31).

refer to the fitting function

$$\theta_{ts} = \frac{\theta_{ts(v=0)}}{1 + 0.00113 \times v^{2.2}} \quad (v \text{ in km/h}) \quad (30)$$

where

$$\theta_{ts(v=0)} = 21.61 + 7.952 \times (\delta_1^\circ - 3) - 0.2 \times (\delta_1^\circ - 3)(9 - \delta_1^\circ) \quad (31)$$

which gives a very good approximation to the numerical results and may be used, together with Eq. (28), to calculate the path curvature depending on  $v$  and  $\delta_1$ . Considering the plot of Figure 3(b), one deduces that quite small path radii may be afforded at low speed with the multi-steering configuration in comparison with the conventional vehicles with just one steering axle. These small radii are attainable with comparatively small steering angles  $\delta_1$  because of the cooperation of the other steering wheels.

The conventional configuration with  $\delta_2 \equiv \delta_3 \equiv 0$  implies that the semi-trailer tail mainly runs inside the tractor path at low speeds. Actually, the sideslip angles tend to vanish for very low speeds and the wheels tend to roll on their planes. Assuming for simplicity that all wheels are ‘condensed’ in the midpoints  $M_i$  of their axles  $i$  and that the six wheels of the semi-trailer are mounted on the same axle ( $3m$ ), and imposing that the normals to the wheel planes converge to the same instant centre of rotation  $C$ , three right-angled triangles may be identified,  $M_1M_2C$ ,  $Q_2M_2C$  and  $Q_2M_{3m}C$ . This permits writing the following

relation between the path radii  $r_1 = M_1 C$  and  $r_{3m} = M_{3m} C$  (kinematical off-tracking):

$$r_1^2 - r_{3m}^2 = (a_1 + a_2)^2 + (b_{2s} + a_3)^2 - (a_2 - b_{2t})^2 \quad (32)$$

As the right side cannot be negative because  $a_2 - b_{2t}$  is much smaller than the other terms, the path of  $M_{3m}$  is inside that of  $M_1$ .

On the other hand, the conventional semitrailers tend to move out of the tractor path at high speeds, because of the centrifugal forces. Therefore, the multi-steering correction must produce discordant and concordant steering of the rear wheels at low and high speed, respectively. Nevertheless, the off-tracking correction is mostly advisable at low speed, as quite small path radii are obtainable for easily manoeuvring inside very narrow spaces. At high speeds, on the contrary, the multi steering configuration gives relatively little advantages in terms of off-tracking correction in comparison with the conventional vehicles, for example when undertaking the gentle curvature of a highway bend. The major concern at high speed must be the control of the radial acceleration to ensure the stability of the articulated vehicle.

#### 4. Entrance and exit from a circular path

The results of the previous section apply to a path with constant curvature but this condition must be preceded and followed by transient periods at the entrance and exit from the curve. Then, one has to search the correct time evolution of  $\delta_2$  and  $\delta_3$  so that the control points  $Q_2$  and  $Q_3$  follow the path of  $Q_1$  exactly during the entrance and exit periods, from zero to  $\delta_{\text{steady}}$  and from  $\delta_{\text{steady}}$  to zero, respectively.

It is here assumed that the changes of  $\delta_1$ , from 0 to  $\delta_{1,\text{steady}}$  and from  $\delta_{1,\text{steady}}$  to 0, do not occur abruptly, due in part to the driver's reaction time and also to the need of preventing any sudden discontinuity and acceleration peak, but take place according to the laws

$$\delta_1 = \frac{\delta_{1,\text{steady}}}{2} \left[ 1 - \cos \frac{\pi(t - t_{0,\text{entry}})}{\Delta t} \right] \quad (\text{entry}) \quad (33)$$

$$\delta_1 = \frac{\delta_{1,\text{steady}}}{2} \left[ 1 + \cos \frac{\pi(t - t_{0,\text{exit}})}{\Delta t} \right] \quad (\text{exit}) \quad (34)$$

where the symbols  $t_0$  indicate the starting times of the driver's manoeuvres,  $\Delta t$  is their duration and Equations (33–34) apply only for  $0 \leq t - t_0 \leq \Delta t$ .

Then, imposing for example  $\Delta t = 1\text{ s}$  and fixing the vehicle speed  $v$  and the input  $\delta_1$  set by the driver, one has to integrate the motion equations numerically by some explicit routine, for example, the Euler-Cauchy method, and has to choose the angles  $\delta_2$  and  $\delta_3$  at each step so to suppress any arising off-track of  $Q_2$  and  $Q_3$ . To this end, it is now necessary to control the motion evolution with respect to the fixed reference  $Oxyz$  and new kinematical equations must be added to the differential system (24):

$$\frac{dx_{Q_1}}{dt} = (u - \omega_t b_1) \cos \theta_t - v \sin \theta_t \quad (35)$$

$$\frac{dy_{Q_1}}{dt} = (u - \omega_t b_1) \sin \theta_t + v \cos \theta_t \quad (36)$$

$$\frac{d\theta_t}{dt} = \omega_t \quad (37)$$

where  $x_{Q1}$  and  $y_{Q1}$  are the coordinates of  $Q_1$ .

Moreover, two sufficiently long arrays are introduced,  $\mathbf{x}_1$  and  $\mathbf{y}_1$ , where the coordinates of the past positions of  $Q_1$  are recorded and updated at each step, to be used as the targets for the following control points. Here, the dimension 510 was chosen for these arrays so that the elementary displacement of each point of the axis, for example of a 17 m long articulated vehicle, may be made as small as  $17 \text{ m} / 510 = 1/30 \text{ m}$ .

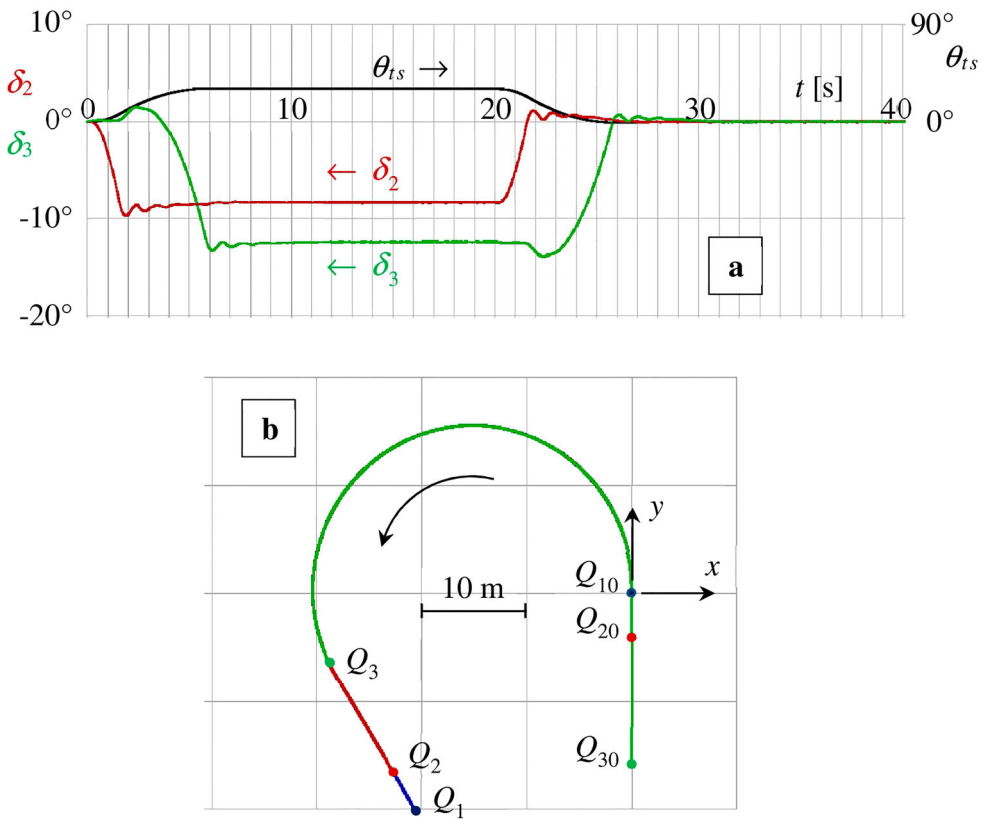
At each step of the integration process, all the elements of the arrays  $\mathbf{x}_1$  and  $\mathbf{y}_1$  are shifted of one place (from position  $j$  to  $j + 1$ ), the last ones are eliminated and the new values of  $x_1(1) = x_{Q1}$  and  $y_1(1) = y_{Q1}$  are calculated using Eqs. (35–36), where the right sides are known by the previous step. Then, the distances of the past positions of point  $Q_1$  from the new one are evaluated until two successive distances are the one a little smaller and the other a little greater than  $b_1 + b_{2t}$ . Hence, a linear interpolation between the coordinates of these two positions gives a very good approximation of the position that must be reached by the second control point  $Q_2$ . An analogous procedure permits to find the position to be reached by the third control point  $Q_3$ , whose distance from  $Q_2$  must be equal to  $b_{2s} + b_3$ .

Then, one integration step is carried out to update  $u, \omega_t, \omega_{ts}, \theta_{ts}$  and  $\theta_t$  by Eqs. (24) and Eq. (37), where the quantities at the right sides are calculated in the previous step, and is repeated iteratively adjusting the angles  $\delta_2$  and  $\delta_3$  until the desired positions of  $Q_2$  and  $Q_3$  are attained. In this process, the tentative positions of  $Q_2$  and  $Q_3$  are calculated by the formulae,  $x_{Q2} = x_{Q1} + (b_1 + b_{2t})\sin\theta_t$ ,  $y_{Q2} = y_{Q1} - (b_1 + b_{2t})\cos\theta_t$ ,  $x_{Q3} = x_{Q2} + (b_{2s} + b_3)\sin(\theta_t - \theta_{ts})$ ,  $y_{Q3} = y_{Q2} - (b_{2s} + b_3)\cos(\theta_t - \theta_{ts})$ , and the successive corrective changes of the steering angles are chosen in the form  $\Delta\delta_2 = d_2/(b_1 + b_{2t})$  and  $\Delta\delta_3 = d_3/(b_{2s} + b_3)$ , where  $d_2$  and  $d_3$  are the transverse distance of the current tentative positions of  $Q_2$  and  $Q_3$  from the desired ones. This choice makes the iterative procedure converge very quickly to the desired steering angles ( $\Delta\delta_2 \rightarrow 0$ ,  $\Delta\delta_3 \rightarrow 0$ ).

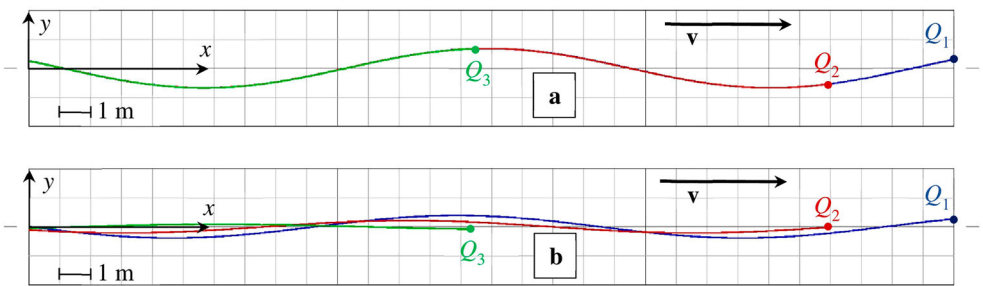
Repeating this procedure in the following integration steps, the proper evolutions  $\delta_2(t)$  and  $\delta_3(t)$  for the off-track cancellation of the control points are readily obtained.

Figure 4(a) shows the angles  $\delta_2(t)$ ,  $\delta_3(t)$  and  $\theta_{ts}(t)$  for the case of an articulated vehicle with the data of Table 1, and the values of  $v$  and  $\delta_1$  in the caption. The entrance into a curved path is imposed for  $t = t_{0,\text{entry}} = 0\text{s}$  and the exit to return to the straight path is imposed for  $t = t_{0,\text{exit}} = 20\text{s}$ . The starting values for  $t = 0$  are:  $u = 0$ ,  $\omega_t = 0$ ,  $\omega_{ts} = 0$ ,  $\theta_{ts} = 0$ ,  $x_j = 0$ ,  $y_j = -(j-1)/30$ ,  $\theta_t = 0$ . The plots are somewhat wavy in correspondence of the changes of the steering angles, but this waviness is rather moderate because of the smoothed entrance and exit connection by Eqs. (33–34). The selected case refers to a path with a small radius and a low vehicle speed and then simulates a slow manoeuvre in a small space, where the off-tracking correction is fundamental and significant improvements are obtainable by the multi-axle steering.

At high speeds, the turning radii cannot but be rather large, to prevent the vehicle rollover, and the corrective steering angles of the back axles are quite small and concordant with the drive axle (see Figure 3). In these conditions, the benefits of the off-tracking cancellation by multi-steering axles are less significant, whereas it is fundamental to limit



**Figure 4.** Numerical results for an optimised run along a bend including the entry and exit phases ( $v = 10$  km/h,  $\delta_1 = 5$  deg). (a) steering angles,  $\delta_2$  and  $\delta_3$ , and angle  $\theta_{ts}$  between the two vehicle units versus time; (b) trajectories of control points,  $Q_1$  (midpoint of tractor front end),  $Q_2$  (centre of fifth wheel coupling), and  $Q_3$  (midpoint of semitrailer rear end).  $Q_{10}$ ,  $Q_{20}$ , and  $Q_{30}$ : starting position.



**Figure 5.** Trajectories of control points,  $Q_1$  (midpoint of tractor front end),  $Q_2$  (centre of fifth wheel coupling), and  $Q_3$  (midpoint of semitrailer rear end). Sinusoidal steering input:  $\delta_1 = \delta_{1\max} \times \sin(2\pi ft)$ . (a) multi-steering vehicle (b) single axle steering vehicle. Data:  $v = 5$  km/h  $\delta_{1\max} = 5$  deg  $f = 0.3$  Hz.

the rearward acceleration peaks arising from sudden steering manoeuvres. It will be shown in the following that the acceleration containment can be easily achieved by the smooth transient phases of Eqs. (33–34).



Figure 4(b) shows the trajectories of the control points,  $Q_1, Q_2, Q_3$ , as they are calculated by the above procedure. It is observable that the paths of  $Q_2$  and  $Q_3$  are exactly superimposed to that of  $Q_1$ , both in the circular part and in the entrance and exit parts.

At last, it is worth noting that it is not convenient to apply the multi-steering technique in the hypothesis of a sinusoidal steering input  $\delta_1(t) = \delta_{1\max} \sin(2\pi ft)$ . This is shown in Figure 5(a,b), where single-steering and multi-steering are compared for the same  $\delta_1$  input. The shift from the straight path for the latter case appears larger on the whole than for the former.

## 5. A formulation for practical use

Examining several cases with different velocities  $v$  and steering angles  $\delta_1$  by numerical integration, it may be observed that some features of the diagrams  $\delta_2(t)$  and  $\delta_3(t)$  recur with remarkable regularity. Taking advantage of these common features, a general-purpose approximate formulation may be outlined, permitting a rapid calculation of the optimal steering angles  $\delta_2(t)$  and  $\delta_3(t)$  without recourse to laborious integration tasks for each new case. This formulation offers a practical tool to programme the steering of all axles along any bent path.

In practice, the results by the 'exact' procedure of the previous section show that the angle  $\delta_2$  begins to vary just after the start of the steering command imposed by the driver and its magnitude increases with the time according to a nearly parabolic law towards that of the steady value  $\delta_{2,\text{steady}}$ , which is approximately reached after the period  $t_{12} = (b_1 + b_{2t})/v$ . It was formerly specified that the steering operation does not start with a sudden step  $\delta_{1,\text{steady}}$ , but is smoothed throughout a time interval  $\Delta t$  according to Eq. (33). As the system does not know in advance the final steering angle  $\delta_{1,\text{steady}}$  desired by the driver at the end of the interval  $\Delta t$ , it cannot guess the corresponding value  $\delta_{2,\text{steady}}$ , whence it is planned to adjust the target angle  $\delta_{2,\text{steady}}$  continuously to the current value of  $\delta_1$  during the connection period  $\Delta t$ , considering it as a temporary steady value  $\delta_{1,\text{steady,temporary}}$ . If  $\Delta t < t_{12}$ , the target value  $\delta_{2,\text{steady}}$  will be reached after the time interval  $t_{12}$ , otherwise it will be reached later, after the interval  $\Delta t$ .

As regards the steering angle  $\delta_3$ , its variation starts immediately after the mentioned period  $t_{12}$ , but two different behaviours appear for the two cases of counter-steering and concordant-steering of the axles 3 and 1 at the steady state (low speeds and high speeds, respectively). In the former case, it is possible to observe a nearly sudden increase of  $\delta_3$  to the approximate value  $-\delta_{2,\text{steady}}(b_1 + b_{2t})/(b_1 + b_{2t} + b_{2s} + b_3)$ , which is concordant with  $\delta_1$ , followed by a nearly parabolic variation towards the value  $\delta_{3,\text{steady}}$ , which is reached after the time interval  $t_{34} = (b_{2s} + b_3)/v$ . Of course, it is preferable to exclude sudden jumps of  $\delta_3$  and thus the approximate formulation schedules a softened change of  $\delta_3$  at the beginning of the interval  $t_{34}$ , similarly to the one described by Eq. (33) for  $\delta_1$ . At high speeds, on the contrary, the steering angle  $\delta_3$  is concordant with  $\delta_1$  and appears to reach its steady value  $\delta_{3,\text{steady}}$  by an approximate parabolic law starting from zero directly.

When the driver operates the return to the straight path, the changes of  $\delta_2$  and  $\delta_3$  occur with nearly parabolic laws again towards the final zero values, in the two successive time intervals  $t_{12}$  and  $t_{34}$ , respectively, and there is a jump of  $\delta_3$  similar to the entrance phase in the case  $\delta_{1,\text{steady}} \times \delta_{3,\text{steady}} < 0$  (low speed).

Recapitulating and introducing the number  $\sigma = 0.5 \times [1 - \text{sgn}(\delta_{1,\text{steady}} \times \delta_{3,\text{steady}})]$  for brevity, where  $\sigma$  may be either 0 or 1, the approximate formulation may be written down as follows.

Entry:

$$\delta_2 = \delta_{2,\text{steady,temporary}}(t) \left( \frac{t - t_{0,\text{entry}}}{t_{12}} \right)^2 \quad \text{for } t_{0,\text{entry}} < t < t_{0,\text{entry}} + \Delta t \quad (38)$$

$$\delta_2 = \delta_{2,\text{steady}} \left( \frac{t - t_{0,\text{entry}}}{t_{12}} \right)^2 \quad \text{for } t_{0,\text{entry}} + \Delta t < t < t_{0,\text{entry}} + t_{12} \quad (\text{only for } \Delta t < t_{12}) \quad (39)$$

$$\delta_3 = \sigma \frac{\delta_{2,\text{steady}}}{2} \left( \frac{t_{12}}{t_{12} + t_{34}} \right) \left[ \cos \frac{\pi(t - t_{0,\text{entry}} - t_{12})}{\Delta t} - 1 \right] \quad (\text{for } t_{0,\text{entry}} + t_{12} < t < t_{0,\text{entry}} + t_{12} + \sigma \Delta t) \quad (40)$$

$$\begin{aligned} \delta_3 = & \left[ \delta_{3,\text{steady}} + \sigma \delta_{2,\text{steady}} \left( \frac{t_{12}}{t_{12} + t_{34}} \right) \right] \left( \frac{t - t_{0,\text{entry}} - t_{12} - \sigma \Delta t}{t_{34}} \right)^2 \\ & - \sigma \delta_{2,\text{steady}} \left( \frac{t_{12}}{t_{12} + t_{34}} \right) \quad (\text{for } t_{0,\text{entry}} + t_{12} + \sigma \Delta t < t < t_{0,\text{entry}} + t_{12} + \sigma \Delta t + t_{34}) \end{aligned} \quad (41)$$

Exit:

$$\delta_2 = \delta_{2,\text{steady}} \left[ 1 - \left( \frac{t - t_{0,\text{exit}}}{t_{12}} \right)^2 \right] \quad \text{for } t_{0,\text{exit}} < t < t_{0,\text{exit}} + t_{12} \quad (42)$$

$$\delta_3 = \delta_{3,\text{steady}} - \sigma \frac{\delta_{2,\text{steady}}}{2} \left( \frac{t_{12}}{t_{12} + t_{34}} \right) \left[ \cos \frac{\pi(t - t_{0,\text{exit}} - t_{12})}{\Delta t} - 1 \right] \quad (\text{for } t_{0,\text{exit}} + t_{12} < t < t_{0,\text{exit}} + t_{12} + \sigma \Delta t) \quad (43)$$

$$\delta_3 = \left[ \delta_{3,\text{steady}} + \sigma \delta_{2,\text{steady}} \left( \frac{t_{12}}{t_{12} + t_{34}} \right) \right] \left[ 1 - \left( \frac{t - t_{0,\text{exit}} - t_{12} - \sigma \Delta t}{t_{34}} \right)^2 \right] \quad (\text{for } t_{0,\text{exit}} + t_{12} + \sigma \Delta t < t < t_{0,\text{exit}} + t_{12} + \sigma \Delta t + t_{34}) \quad (44)$$

The above formulae, Eqs. (38–44), are valid for an articulated vehicle of the type in Table 1. Nevertheless, the steady values of the steering ratios,  $\delta_2/\delta_1$  and  $\delta_3/\delta_1$ , and their correlation with  $\delta_1$  and  $v$ , obviously depend on the geometrical and mechanical characteristics of the particular vehicle under consideration and should be calculated beforehand by the previous numerical procedure. Anyway, the freight and its distribution on the loading bed scarcely affect the above equations but may be found to slightly influence some of the formulae that will be derived below.

The diagrams of Figure 3(a) reported the typical dependence of the angles  $\delta_2$  and  $\delta_3$  on  $v$  for  $\delta_1$  fixed in steady conditions, which dependence reminds the upturned cubic witch of Agnesi that was used in reference [29] for the case of a long two-axle vehicle. Based on this resemblance, two similar formulae are here defined for the correlation between  $\delta_i$  and

$v$  (for  $i = 2$  or  $3$ ), though with non-integer exponents

$$\delta_i = \delta_{i,\infty} - \frac{\delta_{i,\infty} - \delta_{i,0}}{1 - \frac{\delta_{i,0}}{\delta_{i,\infty}} \left( \frac{v}{v_{i,0}} \right)^{m_i}} \quad \text{for } i = 2 \text{ or } 3 \quad (45)$$

Here the exponents to be applied are  $m_2 = 2.4$ ,  $m_3 = 2.2$ , while the other parameters are functions of  $\delta_1$ . The angles  $\delta_{i,0}$  and  $\delta_{i,\infty}$  indicate the values of  $\delta_i$  for  $v = 0$  and  $v \rightarrow \infty$  and the speed  $v_{i,0}$  is the velocity of inversion of the steering angle  $\delta_i$ , i. e. when  $\delta_i = 0$ .

The functions  $\delta_{i,0}(\delta_1)$ ,  $\delta_{i,\infty}(\delta_1)$  and  $v_{i,0}(\delta_1)$  were derived for the particular articulated vehicle with the data of Table 1 and are reported in Figure 6. They may be described by the following matching formulae

$$\frac{\delta_{2,0}}{\delta_1} = -2.191 \quad (\text{nearly constant}) \quad (46)$$

$$\begin{aligned} \frac{\delta_{2,\infty}}{\delta_1} &= 0.365 + (0.582 - 0.365) \times \frac{1 - \exp[-5 \times (\delta_1^\circ - 0.5)]}{1 - \exp[-5 \times 8.5]} \\ &\quad + \frac{(\delta_1^\circ - 0.5)|9 - \delta_1^\circ|^{2.5}}{1000} \end{aligned} \quad (47)$$

$$\frac{\delta_{3,0}}{\delta_1} = -3.029 + (3.029 - 4.173) \times \frac{1 - \exp[0.32 \times (\delta_1^\circ - 0.5)]}{1 - \exp[0.32 \times 8.5]} \quad (48)$$

$$\begin{aligned} \frac{\delta_{3,\infty}}{\delta_1} &= 0.212 + (0.822 - 0.212) \times \frac{1 - \exp[-2.5 \times (\delta_1^\circ - 0.5)]}{1 - \exp[-2.5 \times 8.5]} \\ &\quad + \frac{(\delta_1^\circ - 0.5)|9 - \delta_1^\circ|^2}{2000} \end{aligned} \quad (49)$$

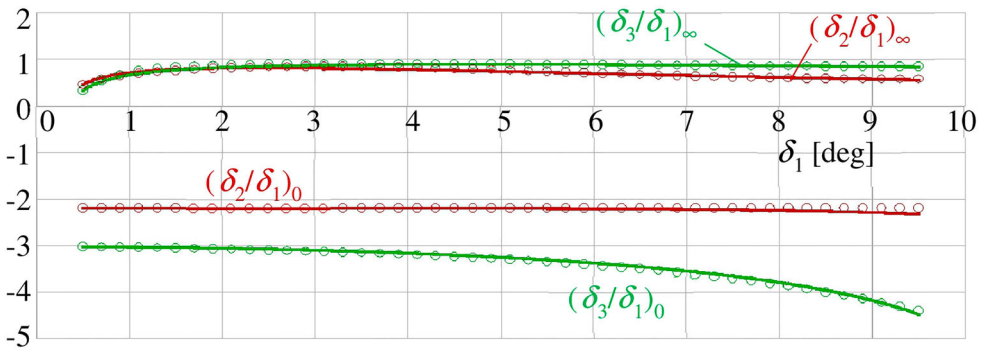
$$v_{2,0}[\text{km/h}] = 35.001 + 1.701 \times \left( 5 - \frac{m_s}{m_t} \right) - \left[ 0.08 - 0.016 \times \left( 5 - \frac{m_s}{m_t} \right) \right] (\delta_1^\circ - 3) \quad (50)$$

$$v_{3,0}[\text{km/h}] = 40.104 + 2.366 \times \left( 5 - \frac{m_s}{m_t} \right) - \left[ 0.766 - 0.133 \times \left( 5 - \frac{m_s}{m_t} \right) \right] (\delta_1^\circ - 3) \quad (51)$$

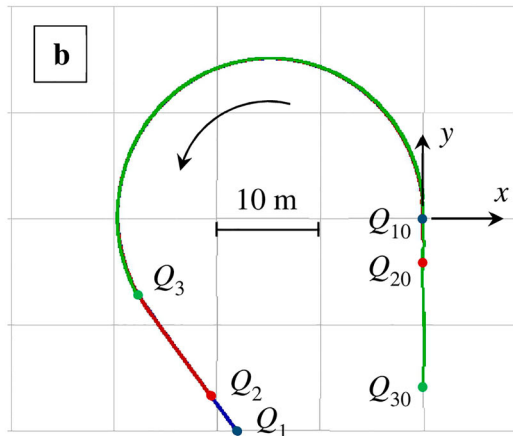
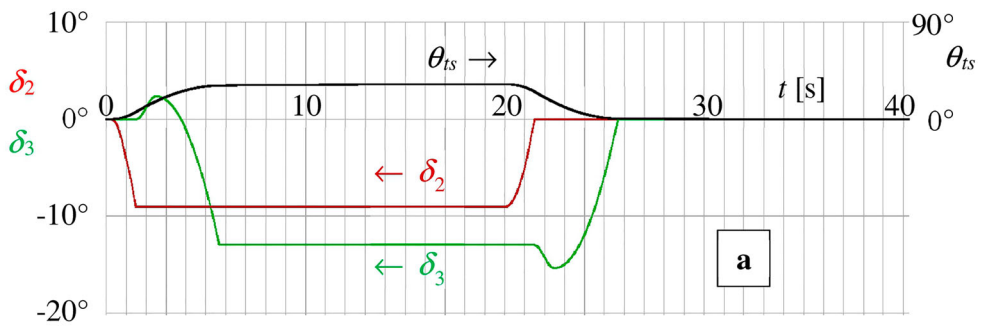
Figure 6 also compares the plots obtainable with the above formulas with the results from the iterative calculation of Section 3. As observable, there is a very good fit with the ‘exact’ results.

Of course, the numerical coefficients appearing in Eqs. (46–51) must be re-calculated *ex novo* for each new vehicle but may be readily used afterwards. Notice the increase of the inversion velocity of Eqs. (50–51) on decreasing the mass  $m_s$  of the semitrailer, which is five times  $m_t$  in full load condition. On the contrary, the angles  $\delta_{i,0}$  and  $\delta_{i,\infty}$  are not influenced in practice by the mass ratio  $m_s/m_t$ , the former because the centrifugal effect become negligible for low speeds and the latter because the steering angles  $\delta_i$  are just small at high speeds and the freight reduction implies smaller order effects.

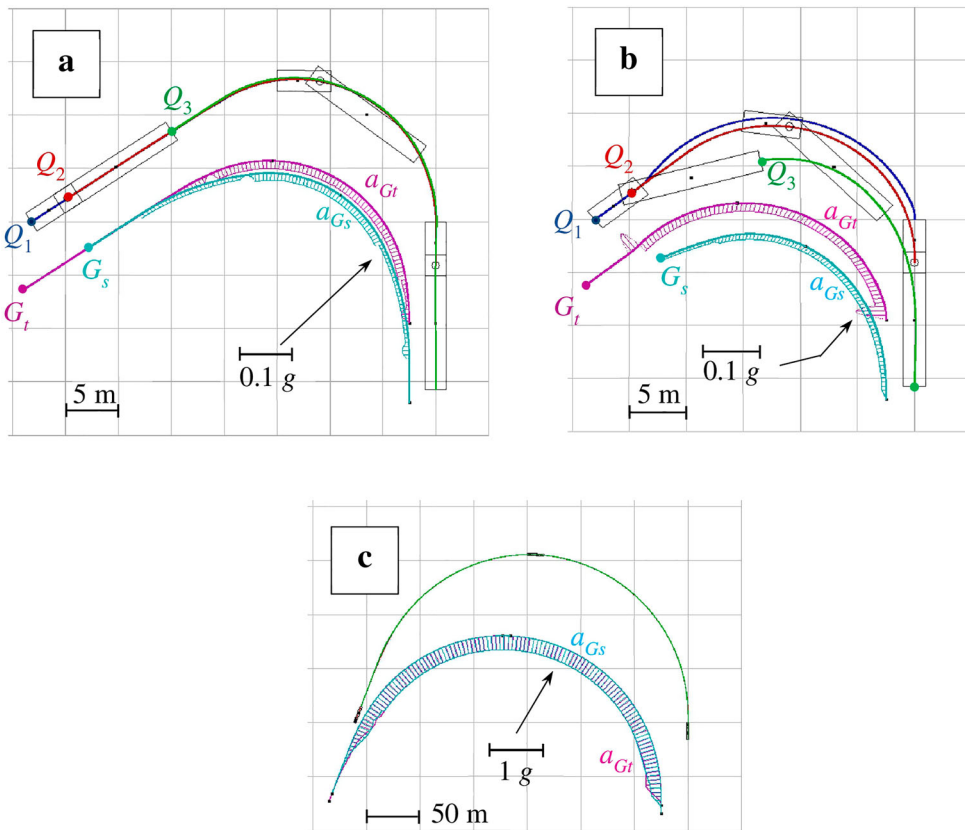
For comparison purpose with Figure 4, Figure 7(a,b) shows the plots of the steering angles and the paths of the three control points  $Q_1, Q_2, Q_3$ , for the same data of Figure 4, but now using Eqs. (38–51). As observable, the three trajectories are still coincident with each



**Figure 6.** Analytical formulae matching the numerical results. Limit values of the ratios,  $\delta_2/\delta_1$  and  $\delta_3/\delta_1$ , for  $v \rightarrow 0$  and  $v \rightarrow \infty$ , as functions of  $\delta_1$ . Continuous line: numerical results; circles: Equations (46–49).



**Figure 7.** Analytical results by matching formulae of Section 5 for an optimised run along a route bend including the entry and exit phases ( $v = 10$  km/h,  $\delta_1 = 5$  deg). (a) steering angles,  $\delta_2$  and  $\delta_3$ , and angle  $\theta_{ts}$  between the two units versus time; (b) trajectories of control points,  $Q_1$  (midpoint of tractor front end),  $Q_2$  (centre of fifth wheel coupling), and  $Q_3$  (midpoint of semitrailer rear end).  $Q_{10}$ ,  $Q_{20}$ , and  $Q_{30}$ : starting position.



**Figure 8.** Results by analytical approximation of Section 5. Figure 8(a,b) refer to slow speed motion and show the trajectories of the control points:  $Q_1$ , midpoint of tractor front end;  $Q_2$ , centre of fifth wheel coupling;  $Q_3$ , midpoint of semitrailer rear end. Figure 8(a) refers to a multi-steering vehicle, whereas Figure 8(b) refers to a single axle steering vehicle with the same path radius of Figure 8a. Various vehicle positions are represented. The trajectories of  $G_t$  (tractor barycentre) and  $G_s$  (semitrailer barycentre) are also shown, shifted downward for clarity reasons, together with the diagrams of the lateral acceleration of the mass centres. Figure 8(c) refers to high speed motion and reports the same quantities of Figure 8(a,b), but with different scales of length and acceleration. The diagrams are nearly indistinguishable. Data:  $g = 9.81 \text{ m/s}^2$ , acceleration of gravity;  $v = 5 \text{ km/h}$  for Figure (a,b);  $v = 70 \text{ km/h}$  for Figure 8(c);  $r = 13.21 \text{ m}$  for Figures 8(a) and 8(b);  $r = 157.53 \text{ m}$  for Figure 8(c);  $\delta_1 = 5 \text{ deg}$  for Figure 8(a,c);  $\delta_1 = 15.82 \text{ deg}$  for Figure 8(b).

other and are nearly the same as in the previous ‘exact’ figure. The trajectory coincidence suggests the feasibility of the off-tracking cancellation by multi-steering systems through very simple feedforward strategies.

Figure 8(a–c) describes the results by the approximate formulation more in detail. Three configurations of the articulated vehicle are examined during the steering manoeuvre and the paths of points  $Q_1$ ,  $Q_2$  and  $Q_3$  are reported, together with the paths of the mass centres and the diagrams of their transverse acceleration, which are shifted downward in the figure for better clarity. Figure 8(a,b) refer to a low speed manoeuvre of an articulated vehicle, with multi-steering and single steering axles, respectively. The multi-steering configuration cancels the path differences completely and realises the optimal movement of the vehicle units.

The off-tracking problem is much less pressing for the case of Figure 8(c), which refers to the case of a vehicle travelling at a higher speed on a bend with gentle curvature, where there is very little difference between multi-steering and single-steering arrangement. Moreover, it is observable how the smoothing of the sudden change of the steering angles during the entry and the exit of the curve by Eqs. (33–34) avoids transverse peaks of the acceleration and maintains it within small fractions of  $g$  throughout the path bend, though it is much larger in Figure 8(c) than in Figure 8(a,b). Better still, the transverse acceleration appears quite low during the steering angle changes. Thus, the rearward amplification effect appears negligible with the described steering technique.

## 6. Conclusions

The behaviour of long vehicles on road reveals important critical aspects that challenge the design engineers long since.

An important undesired drawback of the articulated lorries when undertaking the road bends is the off-tracking of the vehicle tail comparing with the tractor front, consisting in different paths of their midpoints, which may make the trailer invade the contiguous lanes, namely the internal or external lane for low or large speeds, respectively. This unsafe trend is most dangerous at low speed when manoeuvring in small spaces, whereas it is less significant at high speed, where on the contrary it is important to limit the transverse acceleration and prevent rollover. The off-tracking may be successfully contrasted by proper use of multi-steering axles on the various units. It is here shown how specific laws, correlating the steering angles of the axles following the driver-controlled axle with the vehicle speed and the driver's steering command, may be calculated by iterative integration of the motion equations. A simple approximate formulation may be developed for these correlation laws, permitting to handle the steering behaviour of all axles by a convenient feedforward strategy, which is not subject to all those risks of malfunctioning or failure that may occur in the components of a very complex control system. The steering model includes the smoothing of the sudden changes of the steering angles, which turns out to be quite efficient in terms of rearward acceleration reduction and preservation of the rollover stability at high speeds. Moreover, numerical tests prove the vehicle yaw stability for customary loading conditions.

Due to the large path curvature attainable at small speeds and to the centrifugal changes of the cornering forces at high speeds, a nonlinear theoretical model must be used.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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